SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I R.Varshini /192126O207 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 30marks
Understanding of Concepts	25	5
Experimental Design	30	5
Use of Tools	20	5
Analysis of Results	15	5
Documentation	10	5

Total Marks 25 **/30**

Signature of the Faculty

TECHNICAL PRESENTATION QUESTIONS

1. Role of the OSI Model in Software-Defined Networking (SDN)
2. Network Function Virtualization (NFV) and the OSI Architecture
3. Artificial Intelligence for Network Traffic Management
4. Machine Learning-Based Error Detection in Networks
5. Digital Twin Networks and the OSI Model
6. 6G Communication Architecture Compared with the OSI Model
7. Edge Computing and Reliable Data Transmission
8. Cloud Networking Through the OSI Architecture
9. Satellite Internet Communication Using OSI Layers
10. Future of Reliable Networking Beyond the Traditional OSI Model

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable	Basic analysis with limited insights	Poor or incorrect analysis of results

			interpretation		
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

SIMATS ENGINEERING

Intelligence for Network Traffic Management

Applying Artificial Intelligence to keep networks fast, reliable and secure

Presented by

Name: R. Varshini
Reg No: 1921260207
Group No: 08

Course Faculty: Dr. R. Shankar
Course Code: UBA0709

Introduction

Modern networks carry enormous and constantly changing volumes of data from video, cloud apps, IoT devices and mobile users.

Manual, rule-based traffic management cannot keep up with this scale and variability.

Artificial Intelligence lets a network learn from its own data and manage itself.



Growing demand
More users, devices and data every year



Rising complexity
Hybrid, cloud and edge networks



Intelligent answer
AI predicts, decides and adapts

Objectives

SIMATS ENGINEERING

1

Understand AI

Explain what Artificial Intelligence is and its main techniques.

2

Understand traffic

Describe network traffic, its types and key performance measures.

3

Explain the process

Show how AI monitors, analyses and controls network traffic.

4

Assess impact

Identify the benefits, limitations and future scope of the approach.

W hat is Artificial Intelligence?

SIMATS ENGINEERING

Artificial Intelligence (AI) is the ability of machines to perform tasks that normally need human intelligence.

These include learning from data, recognising patterns, predicting outcomes and making decisions.



Machine Learning
Learns patterns from data



Deep Learning
Neural networks for complex data



Reinforcement Learning
Learns by trial and reward



Predictive Analytics
Forecasts future behaviour

W hat is Network Traffic?

SIMATS ENGINEERING

Network traffic is the amount of data moving across a network at a given time, sent as small units called packets.

Common types

- Web browsing
- Video streaming
- Voice and video calls
- File transfer and email

Key measures

- Bandwidth and throughput
- Latency and jitter
- Packet loss
- Congestion level

Traditional vs AI- Driven

SIMATS ENGINEERING

Traditional approach

Fixed, manually written rules

Reacts after problems occur

Slow to adapt to new traffic

Needs constant human effort

AI- driven approach

Learns rules from live data

Predicts and prevents congestion

Adapts in real time

Automates routine decisions

How It W orks

SIMATS ENGINEERING

1. **Collect data** — Capture packets, flows and device logs across the network.
2. **Analyse traffic** — Classify applications and spot patterns or anomalies.
3. **Predict demand** — Models forecast load, congestion and likely failures.
4. **Decide and act** — Reroute, prioritise or limit traffic automatically.
5. **Learn and improve** — Feedback from results keeps refining the models.

Key AI Applications

SIMATS ENGINEERING

Traffic classification

ML identifies which applications generate each flow.

Traffic prediction

Forecasts load so capacity is ready in advance.

Anomaly detection

Flags attacks and unusual behaviour quickly.

Benefits

SIMATS ENGINEERING

Better performance
Lower delay and smoother user experience

Efficient resource use
Balances load and avoids wasted capacity

Stronger security
Early detection of threats

Automation
Less manual configuration and fewer errors

Self- adapting
Responds to change in real time

Reliability
Predicts faults before they cause outages

Future Scope

SIMATS ENGINEERING

Self- driving networks
Fully autonomous configuration and repair

5G, 6G and edge
Real-time AI close to users and devices

IoT and smart cities
Managing billions of connected devices

Predictive security
Stopping attacks before they spread

Explainable AI
Transparent, trustworthy decisions

Intent- based networking
Operators state goals, AI delivers them

Conclusion

SIMATS ENGINEERING

Network traffic is growing faster than manual methods can manage.

AI learns from live data to classify, predict and control traffic.

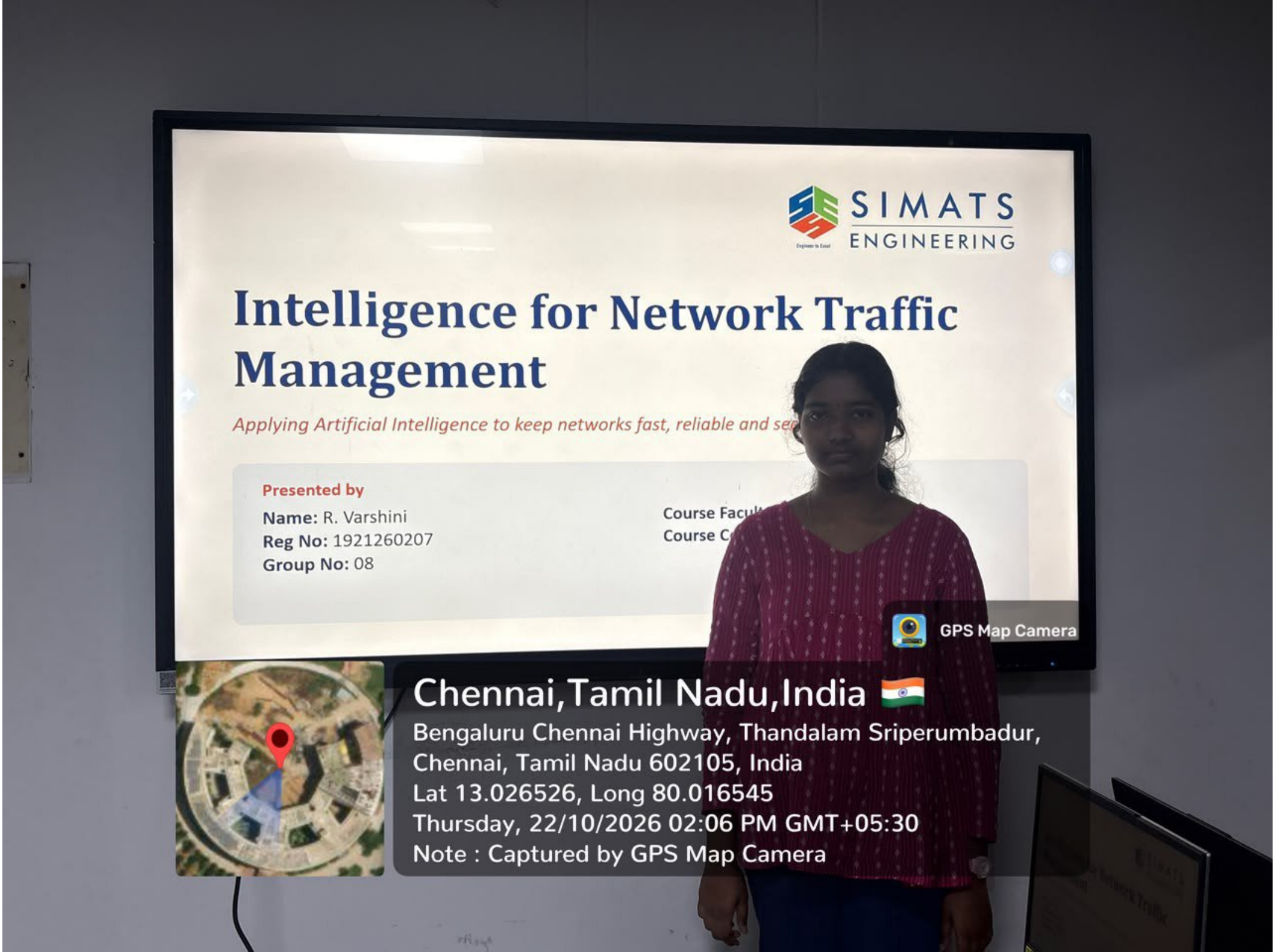
The result is faster, safer and more reliable networks with less effort.

Future networks will increasingly be intelligent and self-managing.

References

1. Russell, S. and Norvig, P., Artificial Intelligence: A Modern Approach, Pearson.
2. Kurose, J. and Ross, K., Computer Networking: A Top- Down Approach, Pearson.
3. Nguyen, T. and Armitage, G. (2008). A survey of techniques for Internet traffic classification using machine learning. IEEE Communications Surveys & Tutorials.
4. Boutaba, R. et al. (2018). A comprehensive survey on machine learning for networking. Journal of Internet Services and Applications.
5. Mao, H. et al. (2016). Resource management with deep reinforcement learning. ACM HotNets.

Thank You


SIMATS
ENGINEERING
Intelligence for Network Traffic Management
Applying Artificial Intelligence to keep networks fast, reliable and se
Presented by
Name: R. Varshini
Reg No: 1921260207
Group No: 08
Course Facul
Course C
GPS Map Camera
Chennai,Tamil Nadu,India
Bengaluru Chennai Highway, Thandalam Sriperumbadur,
Chennai, Tamil Nadu 602105, India
Lat 13.026526, Long 80.016545
Thursday, 22/10/2026 02:06 PM GMT+05:30
Note : Captured by GPS Map Camera